Unkyung Lee

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Professional Summary

- PhD biostatistician and former FDA Mathematical Statistician with expertise in biomedical research, oncology clinical trial design, and regulatory statistical review. Proficient in developing R packages to facilitate the implementation of statistical methods and ensure reproducibility of analyses. Experienced in interdisciplinary collaboration and communicating complex statistical concepts across scientific and regulatory domains.
    - 2 years of experience reviewing pivotal clinical studies for therapeutic and diagnostic devices at the U.S. Food and Drug Administration. Provided written and oral consultations to internal review teams and sponsors, ensuring the application of sound statistical methods to support evidence-based regulatory decisions.
    - 5 years of experience developing statistical methodologies for diverse biomedical studies, including immuno-oncology trial design, competing risks and longitudinal data analyses in HIV and neurodegenerative diseases, and statistical methods for nutritional research.
    - 6 years of experience teaching undergraduate statistics courses to students from diverse backgrounds, including nursing, engineering, the social sciences, as well as student-athletes.

Research Interest

Immuno-oncology Trial Design, Sample Size and Power Calculation, Longitudinal Data Analysis, Competing Risks Data Analysis, Missing Data Analysis, Survival Analysis, Application to HIV, Neurodegenerative Diseases, and Animal Science data.

Experience

Food and Drug Administration, Center for Devices and Radiological Health (CDRH) — Silver Spring, MD
Mathematical Statistician (Remote) — *May 2023 - July 2025*

- Reviewed Pre-submissions, Investigational Device Exemption (IDE) submissions, Pre-market applications including DeNovos and PMAs, primarily for therapeutic general and plastic surgery devices.
- Experienced in reviewing pre-submissions and DeNovos for diagnostic devices such as neurodiagnostic, cardiovascular (AI/ML-based) and cardiac monitoring and abdominal pressure monitoring devices.

Food and Drug Administration, Center for Biologic Evaluation and Research (CBER) — Silver Spring, MD
ORISE Research Fellow (Remote) — *Nov 2020 - Dec 2022*

Mentor: Dr. Zhenzhen Xu

- Developed innovative design and analysis strategy for cancer immuno-oncology (IO) trials involving non-proportional hazards (NPH) patterns.
- Developed the R package PRIME+ for implementation of proposed design and analysis strategy using R and Rcpp.

Texas A&M University — College Station, TX
Post Doctoral Research Associate, NIH T32 Program — *July 2017 - Aug 2020*

Mentors: Distinguished Prof. Raymond J. Carroll & Prof. Tanya P. Garcia

- Developed a location-shift marker model with shape constraints to estimate individual longitudinal trajectories and their inflection points as an objective measure for the onset of Huntington's (neurodegenerative) disease.
- Developed the R package HDChangePoint.
- Provided statistical consultation on analysis of amino acids data produced from Prof. Guoyao Wu's lab in the Department of Animal Science, Texas A&M University.

Sam Houston State University — Huntsville, TX
Lecturer — *Aug 2016 - Jan 2017*

- Taught Introductory Math and Statistics for undergraduate students.

Education

Ph.D., Applied Mathematics, University of North Carolina at Charlotte — Charlotte, NC
Advisor:Prof. Yanqing Sun — *June 2016*

Dissertation: Analysis of Semiparametric Regression Models For the Cumulative Incidence Functions Under the Two-Phase Sampling Designs.

M.S., Mathematics, Ewha Womans University — Seoul, South Korea
Advisor:Prof. Sunyoung Kim — *Feb 2011*

Thesis: Solving a Nonconvex Quadratic Optimization problem by Splitting Coefficient Matrices of Its Constraints and Numerical Experiments.

B.S., Mathematics, Ewha Womans University — Seoul, South Korea
Aug 2007

Publications

Manuscripts in Preparation

1. **Lee, U.** and Xu, Z. Design for immuno-oncology clinical trial with a heterogeneous patient population. ***In progress.***

Peer Reviewed Publications

1. Sun, Y., Heng, F., **Lee, U.** and Gilbert, P.B. (2023). Estimation of conditional cumulative incidence functions under generalized semiparametric regression models with missing covariates, with application to analysis of biomarker correlates in vaccine trials. *Canadian Journal of Statistics*, 51(1), pp.235-25, https://doi.org/10.1002/cjs.11693
2. **Lee, U.**, Carroll, R.J., Marder, K., Wang, Y. and Garcia, T.P. (2020). Estimating disease onset from change points of markers measured with error. *Biostatistics*, 0, 1-17, https://doi.org/10.1093/biostatistics/kxz068
3. **Lee, U.**, Garcia, T.P., Carroll, R.J., Gilbreth, K.R., Wu, G. (2019). Analysis of repeated measures data in nutrition research. *Frontiers In Bioscience*, Landmark, 24, 1378-1390, https://doi.org/10.2741/4785
4. **Lee, U.**, Sun, Y., Scheike, T.H., Gilbert, P.B. (2018). Analysis of generalized semiparametric regression models for cumulative incidence functions with missing covariates. *Computational Statistics and Data Analysis*, 122, 59-79, https://doi.org/10.1016/j.csda.2018.01.003

Selected Contributed Talks & Poster Presentation

1. **U.S. Food and Drug Administration, CBER**, Virtual, Division of Biostatistics Seminar. — Oct 2022
2. **2020 Joint Statistical Meetings (Poster Presentation)**, Virtual Conference. — Aug 2020
3. **Southeast Texas Chapter of the American Statistical Association Poster Session**, Texas A&M University, College Station, TX. — April 2019
4. **2019 Eastern North American Region (ENAR) Annual Meeting**, Philadelphia, PA. — March 2019
5. **Statistical Bioinformatics and Cancer Symposium (Poster Presentation)**, Institute for Applied Mathematics and Computational Science, Texas A&M University, College Station, TX. — Sept 2018
6. **20th Meeting of New Researchers in Statistics and Probability**, Simon Fraser University, Burnaby, Canada — July 2018

Teaching Experience

Instructor at Department of Statistics, Texas A&M University — College Station, TX
2019-2020

1. Summer 2020: STAT 642 Method of Statistics II - Graduate Course on Design and Analysis of Experiment
2. Fall 2019: STAT 302 Statistical Methods - Introductory of Statistics

Instructor at Department of Mathematics and Statistics, Sam Houston State University — Huntsville, TX
2016 - 2017

1. Fall 2016: MATH 1314 Pre-Calculus Algebra
2. Fall 2016: STAT 3379 Statistical Methods in Practice with SPSS

Instructor at Department of Mathematics and Statistics, UNC at Charlotte Charlotte, NC
2013-2016

1. Spring 2016: STAT 1221 Elements of Statistics I
   - Introductory of Statistics
2. Summer 2015: STAT 2122 Introduction to Probability and Statistics
   - Introductory of Statistics, required calculus prerequisite
3. Spring 2014: STAT 1222 Introductory Statistics
4. Fall 2013: MATH 0900 Math Study Skills & Algebra Review

Professional Activities

1. Attended 2024 Advanced Medical Technology Association conference, Washington D.C. April 2024
2. Served as Journal Referee, Biostatistics Dec 2022
3. Served as Session Chair at the 20th Meeting of New Researchers in Statistics and Probability, Simon Fraser University, Burnaby, Canada July 2018
4. Attended Workshops:
   (a) P.O.W.E.R Academic Writing Workshop at Texas A&M University, College Station, TX June 2019
   (b) International Symposium on Nutrition & Human Health, Tohoku University (Sendia, Japan) and Texas A&M University at College Station, TX Nov 2018
   (c) Jump-Start Your Writing Productivity at Texas A&M University, College Station, TX Jan 2018

Selected Honors and Awards

1. **Two group review awards** for the orthopedic device and the cardio-physiological monitoring device. 2024-2025
2. **A group project award** for dermal fillers worked on during 2023 - 2024. Oct 2024
3. Travel Award for Young and Minority Researchers, 6th Workshop on Biostatistics and Bioinformatics, Georgia State University, Atlanta, GA. May 2018
4. Graduate Teaching Award, In recognition of excellent undergraduate teaching in the Department of Mathematics and Statistics, University of North Carolina at Charlotte. 2014-2015
5. Brain Korea 21 Fellowship, National Research Foundation, Korea. 2008-2010

Statistical Computing and Software

- **Proficiency**: R, Rcpp, Matlab
- **Basic**: SAS, C, SPSS